



# Simulation-based insights into community uniqueness within fragmented landscapes

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## Abstract

**Context** Local contribution to beta-diversity (LCBD) assesses community composition uniqueness of sites within a region. While it is useful to identify sites with exceptional species composition and, thus, prioritize conservation actions, it is unclear what determines community uniqueness in patchy habitats. **Objectives** The goal of this study was to clarify the processes underlying community uniqueness in fragmented landscapes and understand how habitat characteristics and community characteristics affect this beta-based diversity indicator.

**Methods** We simulated neutral metacommunities and used a variance-based method to assess the

contribution of each habitat patch to total beta-diversity, both in terms of replacement and abundance difference. Then, we analyzed the effects of patch and metacommunity characteristics on LCBD.

**Results** Community uniqueness in species replacement and richness/abundance differences responded differently to community and patch features. Patch quality was the habitat attribute with the strongest effects on all community uniqueness aspects, leading to singular assemblages with high species richness and abundance of rare species. While patch connectivity promoted singular assemblages with high richness, patch size increased community uniqueness in species replacement over time, favoring assemblages with high abundances of rare species.

**Conclusions** Community uniqueness in species replacement and richness/abundance differences convey different information and should be considered separately to propose adequate conservation strategies. Habitat quality emerged as a critical factor in shaping beta-diversity, suggesting that it should be a primary focus of conservation efforts. Future studies are needed to evaluate the generality of our results in different spatial and ecological contexts.

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## Introduction

One of the primary causes of biodiversity declines is habitat change (Betts et al. 2022; Tan et al. 2022). Fueled by human actions and climate change, it has become a global issue affecting all biomes and ecosystems (Bellard et al. 2012) and is anticipated to gain even more momentum in the coming years (Jones and Driscoll 2022). The devastating effects of habitat alteration on biodiversity are mediated by the structural changes it incurs in landscapes that, in turn, produce profound and long-lasting effects on population and community functioning (Wang et al. 2021). With growing concern about habitat loss and fragmentation, much research has been devoted to the regulation of species diversity in fragmented ecosystems (Steffan-Dewenter 2003; Peay et al. 2007; Fahrig 2013; Wulf and Kolk 2014; Hanski 2015; Herrault et al. 2016; Rybicki et al. 2020); and has provided overwhelming evidence that species richness within habitat fragments increases with patch size and connectivity among sites (Turner and Gardner 2015). While there is ongoing debate about the underlying processes (De Sanctis et al. 2010; Hilty et al. 2012; Joern and Laws 2013), patch size, connectivity relationships and corresponding species richness have become a cornerstone in landscape ecology. As a result, large and well-connected patches are often given high conservation priority (Ovaskainen 2002; Wintle et al. 2019). However, such conservation strategy, aiming at preserving sites with the highest richness levels, might not be optimal for protecting a maximum number of species (Justus and Sarkar 2002; Hillebrand et al. 2018). This is because species rich habitats do not necessarily provide a better representation of the total species pool present in fragmented landscapes than sets of other habitat patches with high complementarity (i.e., that have different species) (Wintle et al. 2019; Hill et al. 2021). In fact, recent literature has highlighted the importance of sites with low richness in that respect, which often hold the most unique community composition (Legendre and De Cáceres 2013; García-Navas et al. 2022). To clarify the usefulness of patch traits for guiding conservation actions, it is thus necessary to consider their effects on diversity indicators with more informative value for species conservation at the landscape scale than species richness.

The local contribution to beta-diversity (LCBD) is a recently proposed indicator that measures the degree of uniqueness of sites present in a region in terms of community composition (Legendre and De Cáceres 2013). Based on the partitioning of beta-diversity, this metric quantifies the extent to which each site contributes to the regional variation in species' identities and abundances; and is well-suited to identify sites that are exceptional in that sense. Both theoretical and empirical considerations suggest a determinant role of patch characteristics on the distribution of community uniqueness in fragmented landscapes. There is a general acceptance that patch features are key drivers of beta-diversity in fragmented landscapes (Si et al. 2015; Edge et al. 2017; Gavish et al. 2019; Zhao et al. 2021) and mounting evidence demonstrating their influence on community composition (Mazerolle et al. 1999; Husté and Boulmier 2011; Dale 2019; Steel et al. 2022). However, evidence for patch characteristics' effects on community uniqueness is scarce and inconsistent, with studies suggesting some effects of patch size and connectivity (Wintle et al. 2019) and others that did not find such evidence (Yan et al. 2021; Lozada and Bertin 2022).

More investigations on the distribution of LCBD in fragmented habitats are thus necessary to clarify how patch characteristics influence biodiversity patterns across landscapes. However, disentangling their general effects might be a difficult task in natural conditions as multiple factors combine to explain community uniqueness and vary in importance depending on the context (Hill et al. 2021; Santos et al. 2021; Yao et al. 2021). Simulation modelling approaches can help solve this issue by incorporating mechanisms for patch traits to drive community uniqueness in the absence of interfering factors. While a number of simulation studies have explored the impacts of habitat characteristics (e.g., size, isolation) on alpha diversity (Kadmon and Allouche 2007; Thompson et al. 2017; Laroche et al. 2020), none have actually investigated their effects on community uniqueness. The main goal of our study was to assess these effects and involved mechanisms. Using a spatial neutral model to simulate metacommunities, we analyzed the distribution of community uniqueness in fragmented landscapes and disentangled the underlying processes by investigating local contribution to total beta-diversity and to its two additive components: species

replacement and richness/abundance differences (Legendre 2014). Given that these components result from different driving forces, they provide insightful information for understanding diversity patterns (Baselga 2010; Legendre and De Cáceres 2013). Species replacement, which refers to situations where species replace each other at new sites (Chase and Myers 2011), is essentially caused by environmental filtering, competition and historical constraints (Qian and Ricklefs 2007; Baselga 2010; Leprieur et al. 2011). It is usually the dominant component of beta-diversity in landscapes characterized by large environmental variations (Angeler 2013; Nunes et al. 2016; Mateo-Tomás et al. 2019). Whereas differences in local species richness (i.e., when using species occurrence data) or richness and abundances (i.e., when using species abundance data) are usually driven by differences in niche diversity and carrying capacity at different locations (Legendre 2014).

Considering the absence of environmental filtering in our metacommunity model, we made two main predictions. First, we expected species replacement to contribute little to beta-diversity due to the absence of environmental effects and thus anticipated richness/abundance differences to dominate. Second, we predicted that, as strong determinants of community size and persistence (Pardini et al. 2005), patch size, connectivity and quality will strongly influence community uniqueness in terms of losses and gains in local species richness and abundances. To evaluate these predictions, we first examined which beta-diversity components dominate in neutral communities. Then, we analyzed how community and patch characteristics influence the distribution patterns of community uniqueness in patchy habitats. And, finally we assessed whether those effects are similar for uniqueness in overall community composition and in terms of species replacement and community abundance.

## Materials and methods

### Metacommunity simulation model

We simulated biological communities in  $50 \times 50$  grid cell landscapes including randomly distributed habitat cells (i.e., patches) plus a regional pool located equidistant from all habitat patches. The number of habitat patches was arbitrarily set to 100.

All remaining cells formed an inhospitable matrix. Values for parameters associated with demographics and patch characteristics were selected following model calibration to identify parameter combinations that provided a stable total abundance after 100-time steps, and were also realistic (Appendix S1: Fig. S1). The goal of this calibration was to provide a stable simulation environment in which we could test the key processes of interest associated with beta-diversity. Each habitat patch was characterized by a randomly assigned specific size and quality with the aim of achieving moderate variation between patches (coefficient of variation = 0.4). The variable patch size was established as the maximum carrying capacity for each site and distributed according to the Normal distribution with mean of 1500 individuals and standard deviation of 600 individuals (Table 1). Habitat quality was estimated by assigning random rates of mortality to each patch that were distributed as Normal with mean 0.3 and standard deviation of 0.12 (Table 1). In total, 50 replicate landscapes were generated with a different random placement of the sites and assignment of patch size and mortalities in each of the landscapes. Descriptive statistics of patch attributes within each replicate landscape are provided in Appendix S1: Table S1.

Across the 50 landscapes, we simulated 600 generations using CDMetaPOP v1.43 (Landguth et al. 2017), which is a spatially-explicit individual based metapopulation simulation program that simulates demographic processes including growth, reproduction, dispersal and mortality. Modifications to the existing CDMetaPOP framework were added such that multiple species and metacommunity dynamics could be simulated. Specifically, we allowed for simulations of species by assigning a species identifier from 1 to 50 to each individual and only allowed mating to occur within species. At each time step, demographic processes were simulated within each habitat patch using an age-based class structure and balanced sex-ratio. We included four age classes and varied mortality and reproduction parameters for each class (Table 1). The youngest class, experiencing a moderate mortality rate, consisted of individuals that were not yet able to reproduce. The reproductive stage spanned over two adult classes, with each experiencing low mortality rates and considering a fixed number of descendants per female and time-step.

**Table 1** Summary of main parameters used for the metacommunity simulations

| Type        | Parameter                                                               | Source                    | Patch habitats                                                                                          |
|-------------|-------------------------------------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------|
| Demographic | Number of age class                                                     | 4                         | 4                                                                                                       |
| Demographic | Sex ratio at initialization (M:F)                                       | 1:1                       | 1:1                                                                                                     |
| Demographic | Mortality rate per age class in percentage                              | 0, 0, 0, 0                | 0.10, 0.05, 0.05, 0.80                                                                                  |
| Demographic | Fecundity per age class                                                 | 0, 3, 3, 0                | 0, 2, 2, 0                                                                                              |
| Habitat     | Carrying Capacity (Number of individuals)                               | 75,000                    | Randomly assigned from a normal distribution with a mean 1500 individuals and standard deviation of 600 |
| Habitat     | Number of individuals at initialization                                 | 75,000 (1500 per species) | 0                                                                                                       |
| Habitat     | Mortality rate per patch                                                | 0                         | Randomly assigned from a normal distribution with a mean of 0.30 and standard deviation of 0.12         |
| Habitat     | Relative dispersal probability between patches after colonization event | NA                        | $1/(10 \text{ (Euclidean Distance)}^2)$                                                                 |

Source = the initial starting source patch ( $n = 1$ ); Habitat = all other patches ( $n = 100$ )

The senescent class, with the highest mortality rate, included individuals that had ceased reproducing.

The metacommunity was initialized in each landscape during the first 15 time-steps, during which the regional pool, formed by 50 species with 750 females and 750 males per species at initialization, served as a source of dispersal to the habitat patches. During the dispersal process, which occurred at each time step, individuals could either move to another habitat patch or stay in their home patch. Dispersal was allowed between all pairs of patches, with the probability of individual dispersal between any two patches set as an inverse function of Euclidean distance between them (Table 1). After initialization, dispersal only occurred among habitat patches.

#### Beta-diversity estimates and community uniqueness

All statistics were conducted in R (R Core Team 2021). Following a 150 time-steps burn-in period, we analyzed beta-diversity at every 50 time-steps. Thus, we estimated metacommunity metrics for 500 patch-by-species abundance matrices produced for each landscape replicate (i.e., 50 in total) and investigated time-step (i.e., 10 in total). To calculate and partition beta-diversity, we used the variance-based approach of Legendre and De Cáceres (2013). Total beta diversity ( $BD_{Tot}$ ) was estimated as the total variance of the community composition table from a dissimilarity matrix ( $D$ ) produced using the Ružička

index. This measure is the quantitative form of Jaccard's dissimilarity, and is thus appropriate to estimate community dissimilarity from non-transformed species abundance data (Legendre 2014). To partition total-beta diversity into its additive components (i.e.,  $BD_{Repl}$  and  $BD_{AbDiff}$ ),  $D$  was decomposed into the replacement ( $D_{Repl}$ ) and richness/abundance difference ( $D_{AbDiff}$ ) similarity matrices using the *beta.div.comp* function of the 'adespatial' R package (Dray et al. 2020).

Community uniqueness was assessed by calculating the local contribution of each patch to beta-diversity (LCBD) by applying the 'adespatial' *LCBD.comp* function on each dissimilarity matrix. Thus, we obtained three community uniqueness measures corresponding to  $LCBD_{Tot}$ ,  $LCBD_{Repl}$ ,  $LCBD_{AbDiff}$ , which assess the specific contributions of each habitat patch to  $BD_{Tot}$ ,  $BD_{Repl}$ ,  $BD_{AbDiff}$ , respectively.  $LCBD_{Tot}$  is thus a measure of uniqueness in overall community composition, while  $LCBD_{Repl}$  and  $LCBD_{AbDiff}$  measure uniqueness in terms of species replacement and community richness and abundance differences, respectively.

#### Predictive variables

##### *Metacommunity variables*

For each landscape replicate and focal time-step, we investigated effects of habitat patch community

characteristics on community uniqueness. The community parameters considered included the within-patch total species abundance, richness and evenness, and the relative abundance of rare species. To estimate species evenness, we used Pielou's evenness, which was calculated as follow  $J = -\sum [p_i * \log(p_i)] / \log S$ , where  $p_i$  is the proportion of species  $i$  and  $S$  is the number of species. We considered as rare species those occurring in less than 40% of patches (Legendre and De Cáceres 2013; Yao et al. 2021). In addition, we computed for each species the number of sites that it occupied and its total abundance in the landscape.

### *Habitat patch characteristics*

The patch habitat characteristics used as explanatory variables for the LCBD analyses were habitat size, quality and connectivity. The habitat size and quality attributes of each patch (defined above) were patch maximum carrying capacity and mortality. They were both established during the landscape construction process (see above). To estimate patch connectivity, we calculated the immigration probability of each patch by summing the dispersal probability of the focal patch with all the others.

### *Data analysis*

As LCBD assumed values between 0 and 1, we investigated the effects of patch and metacommunity characteristics on community uniqueness by carrying out beta regression models, which allowed modeling bounded response variables (Cribari-Neto and Zeileis 2010). These regressions were conducted using the 'betareg' R package (Cribari-Neto and Zeileis 2010). LCBDs were modeled in relation to local community characteristics (species abundance, richness, evenness, and relative abundance of rare species in the patches and their quadratic terms) and habitat variables (patch connectivity, size and quality).

To avoid overfitting and collinearity issues, we first performed a pre-selection of the predictors that most often contributed to LCBD throughout the landscapes and time-steps. This step required us to model the beta-diversity contribution metrics using a search strategy similar to that carried out by the *regsubset* function of the 'Leaps' package (Lumley 2017) in R. These analyses were conducted for each landscape and focal-time step (i.e., 500 regressions for each

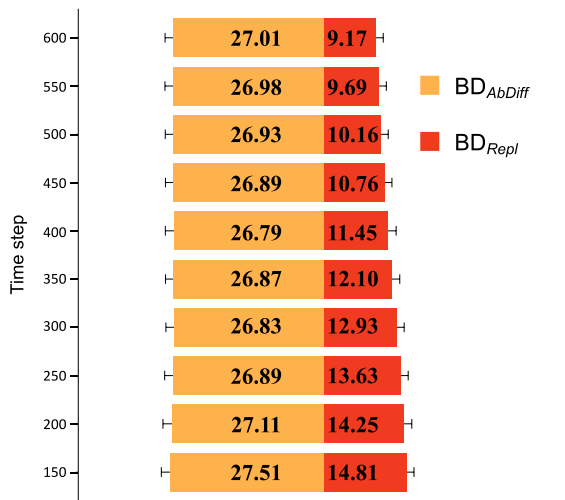
beta-diversity contribution metrics). For each of them, the best fitting model with a given number of predictive variables (i.e., one to 11) was identified using the corrected Akaike's Information Criterion (AICc) (Burnham and Anderson 2002). Then, we compared the performance of these models and retained the one leading to the lowest AICc with no variance inflation factors exceeding 5. Variables occurring in more than 40% of the cases were retained and used in the final multiple regression analyses of the LCBD estimates across all landscapes and time steps. To allow comparisons of their effects, the predictor variables were standardized with a mean of zero and a standard deviation of one before conducting the regressions. Significance of the predictor effects was assessed using the associated 95% confidence intervals. The independent and non-independent effects of patch and species variables on the LCBD estimates were investigated through variation partitioning (Legendre and Legendre 2012).

## **Results**

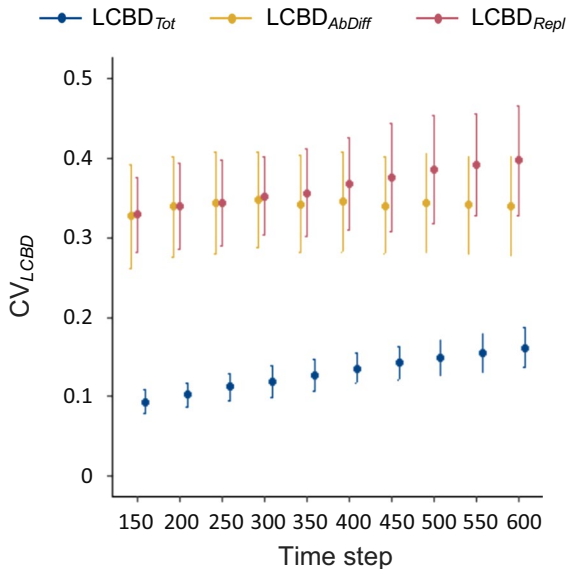
### *Beta diversity and community uniqueness*

The simulated meta-communities displayed high levels of total beta-diversity ( $BD_{Tot}$ ). The average  $BD_{Tot}$  values across all landscapes were 0.42 ( $SD \pm 0.001$ ) and 0.36 ( $SD \pm 0.001$ ) for time-step 150 and 600, respectively (Fig. 1, Appendix S1: Table S2) compared to a maximum possible value of 0.5 with the Ružička dissimilarity index (Borcard et al. 2018). The richness/abundance difference component ( $BD_{AbDiff}$ ) largely dominated across all time-steps (Fig. 1). It accounted on average for 65–75% of  $BD_{Tot}$ . And, while  $BD_{AbDiff}$  remained relatively stable over time,  $BD_{Repl}$  slowly but steadily decreased from time step 150–600 (Fig. 1, Appendix S1: Table S2).

The patterns of community uniqueness contrasted greatly between  $BD_{Tot}$  and its components. The coefficients of variation (CVs) of patch contributions to total beta-diversity averaged 0.13 across all landscapes and rarely exceeded 0.20 (i.e., for only 1.4% of the 500 replicates, Appendix S1: Table S3, Fig. 2). Such moderate CVs indicate that all the patches of the landscape contributed relatively evenly to  $BD_{Tot}$ . In contrast, much higher levels of variation were detected in the site contributions to  $BD_{Repl}$



**Fig. 1** Average beta-diversity across the 50 simulated meta-communities ( $BD_{Tot}$ ) and its partitioning into the species replacement ( $BD_{Repl}$ ) and richness/abundance difference components ( $BD_{AbDiff}$ ). The error bars represent the standard deviations of the beta-diversity components. Average  $BD_{Tot}$  is the sum of average  $BD_{Repl}$  and  $BD_{AbDiff}$



**Fig. 2** Degrees of variation among patches contributions to beta-diversity. The dots and error bars represent the mean and standard deviations of the coefficients of variation (CV) calculated for the local contribution to total beta-diversity estimates (i.e., local contributions to beta-diversity— $LCBD_{Tot}$ —and to its species replacement— $LCBD_{Repl}$ —and richness/abundance difference components— $LCBD_{AbDiff}$ )

and  $BD_{AbDiff}$ . The average CVs of  $LCBD_{AbDiff}$  and  $LCBD_{Repl}$  were three times higher than those displayed by  $LCBD_{Tot}$  (Fig. 2) ranging between 0.18 and 0.52 for  $LCBD_{AbDiff}$  and between 0.24 and 0.64 for  $LCBD_{Repl}$ .

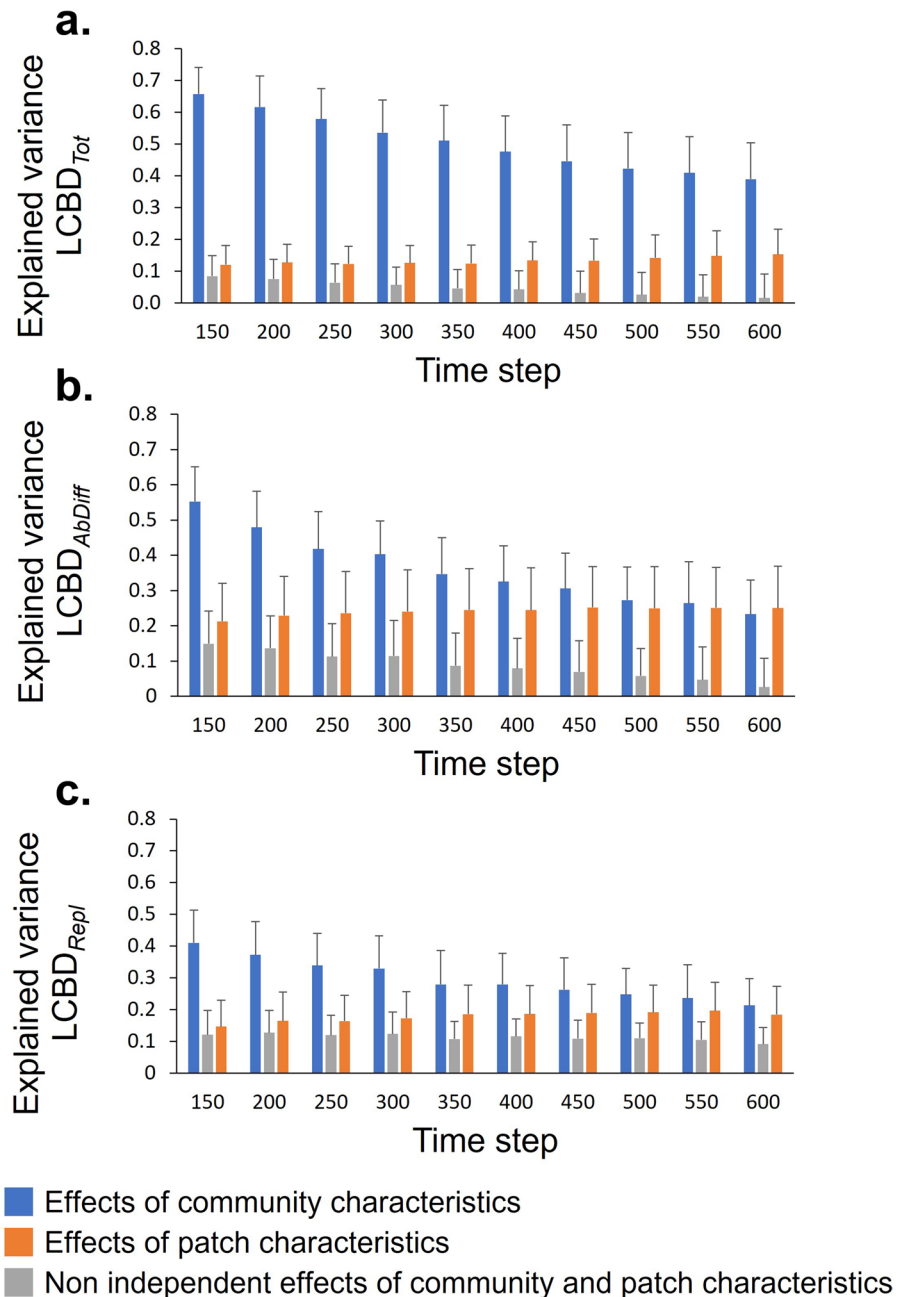
### Determinants of community uniqueness

Both community and patch characteristics were retained as explanatory variables for the beta regressions of the community uniqueness parameters (i.e., LCBDs, Appendix S1: Table S4). The levels of variance they accounted for were higher for  $LCBD_{Tot}$  and  $LCBD_{AbDiff}$  (i.e., 53–69% vs 46–62%, respectively, Appendix S1: Table S5) than for  $LCBD_{Repl}$  (i.e., 31–44%). The relative effects of the community and patch characteristics varied depending on the LCBD components. The community characteristics had, at all time-steps, a predominant influence on uniqueness in overall community composition, explaining up to 65% of the spatial variation in  $LCBD_{Tot}$  (Fig. 3a). In contrast, the patch and community characteristics provided, in the last time steps, relatively similar explanatory power in community uniqueness in terms of richness/abundance difference and species replacement (Fig. 3b, c). The non-independent effects between community and patch characteristics tended to be low, except for  $LCBD_{Repl}$ , for which it accounted for about 30% of the total explained variance.

While the three LCBD indices related significantly to the same sets of community characteristics, none displayed similar relationship patterns (Fig. 4, Appendix S1: Fig. S2). Sites with more unique community compositions ( $LCBD_{Tot}$ ) were sites exhibiting low richness and evenness but high total abundances and abundances of rare species (Appendix S1: Table S6, Fig. 4). The most unique sites in terms of richness/abundance difference were also sites with low richness levels and high species abundances but presenting high evenness and low abundances of rare species (Appendix S1: Table S6, Fig. 4). As for the most unique sites in terms of species replacement, they occurred in sites with completely opposite community characteristics than those presenting high  $LCBD_{AbDiff}$  (Appendix S1: Table S6, Fig. 4).

Site mortality was the patch characteristic that most strongly influenced community uniqueness and

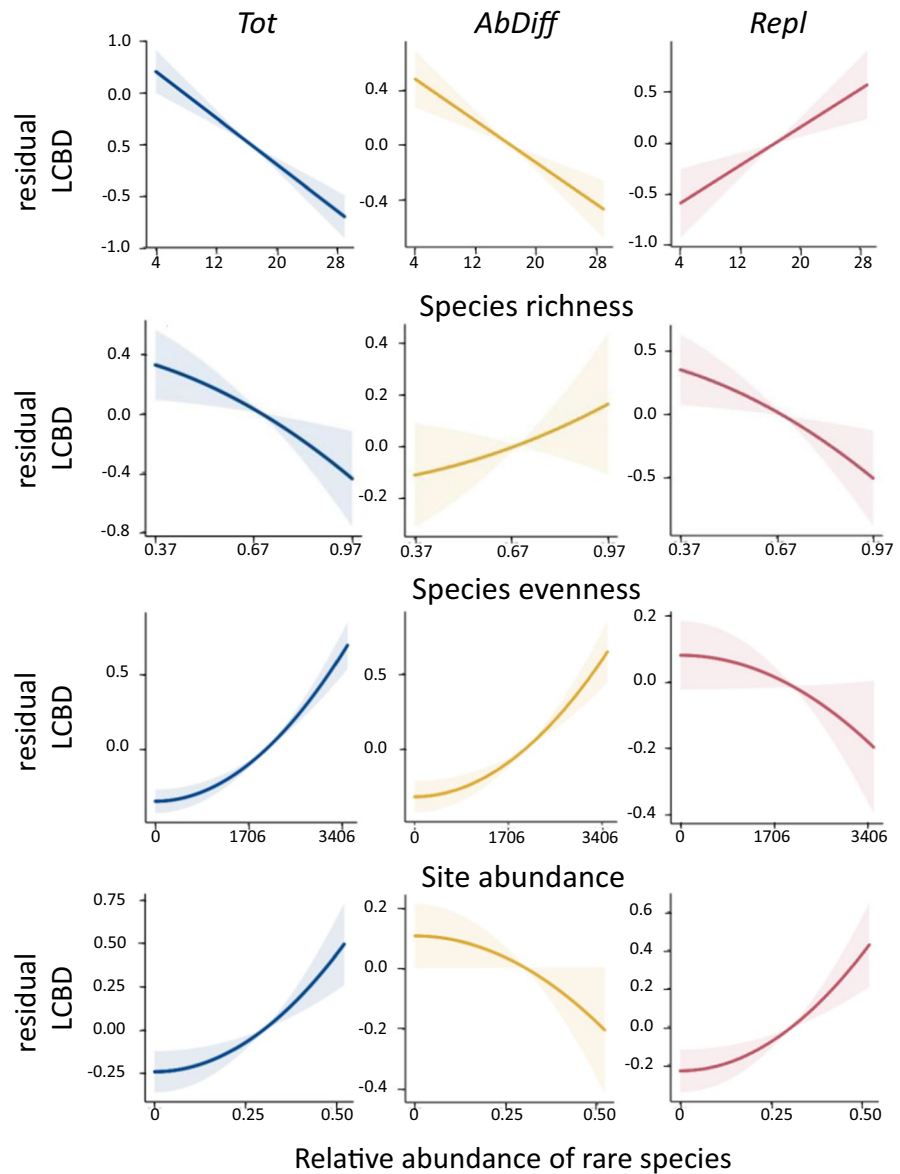
**Fig. 3** Variation partitioning of the effects of local community (e.g., species richness, quadratic Pielou's evenness, quadratic site abundance, quadratic relative abundance of rare species) and patch characteristics (e.g., patch connectivity, carrying capacity and mortality) on local contributions to total beta-diversity ( $LCBD_{Tot}$ ) and its species replacement ( $LCBD_{Repl}$ ) and richness/abundance difference components ( $LCBD_{AbDiff}$ ). Bars represent the fractions of the average variance (pseudo- $R^2$ ) in the  $LCBD$  estimates associated to effects of the community and habitat variables and to their non-independent contributions. Error bars display the standard deviations



its components (Appendix S1: Fig. S2, Appendix S1: Table S7). While it increased community composition uniqueness and the richness/abundance difference component, it had an opposite effect on species replacement uniqueness (Appendix S1: Table S7, Fig. 5). Overall, the detected patch effects revealed that more unique community compositions arose in small and isolated patches and subjected to high

mortality rates (Appendix S1: Table S7, Fig. 5). However, community uniqueness in richness/abundance difference only increased with patch isolation and mortality but did not relate to site carrying capacity. In contrast, uniqueness in species replacement showed higher levels of uniqueness in sites with low mortality rates and responded positively

**Fig. 4** Predicted partial effects of the patch level community characteristics on local contributions to beta-diversity ( $LCBD_{Tot}$ ) and its species replacement ( $LCBD_{Repl}$ ) and richness/abundance difference components ( $LCBD_{AbDiff}$ ) for step time 600. The solid line corresponds to the average predictions across the 50 landscape replicates and the shaded areas to the mean  $\pm$  SD intervals



to patch size in the last time steps (Appendix S1: Table S7, Fig. 5).

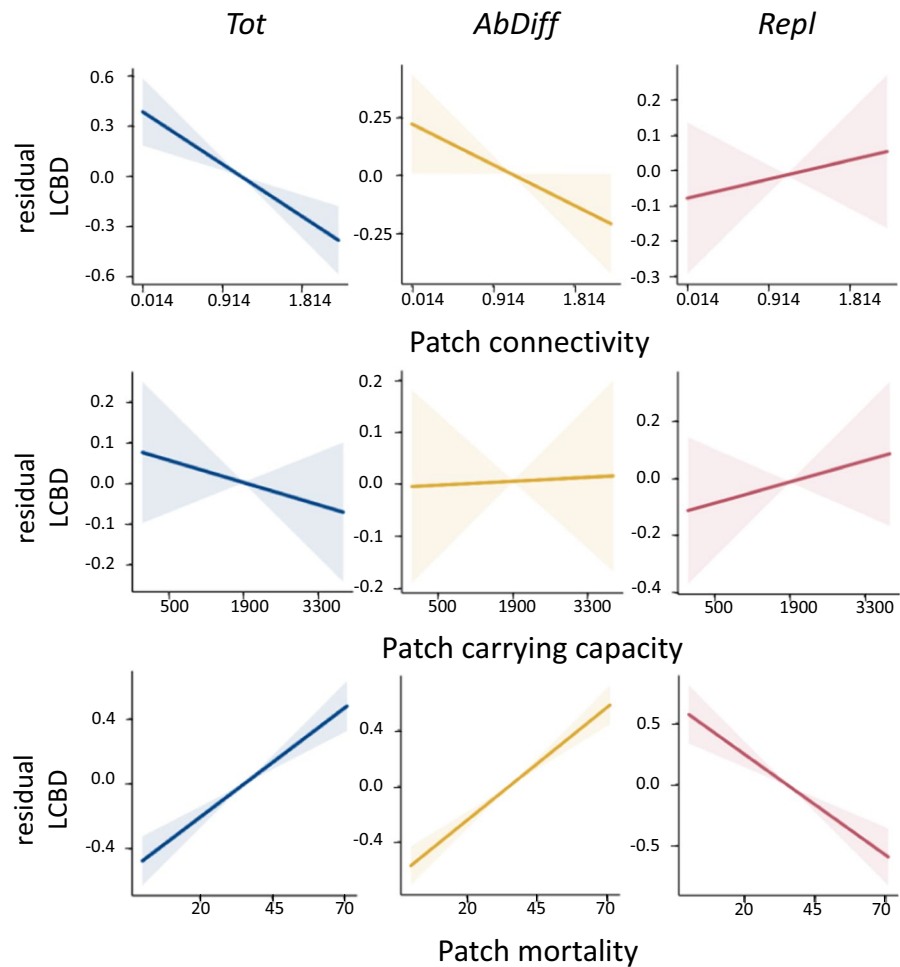
## Discussion

Mechanisms driving beta-diversity in fragmented landscapes

Based on a neutral metacommunity model, our study provides baseline expectations on the structuring of

beta-diversity in fragmented habitats. Overall, we found strikingly high levels of beta-diversity, which is in line with theory and empirical research highlighting the potential for stochastic processes to generate spatial variation in species assemblages (Vellend et al. 2014; Zhang et al. 2016). The fact that beta diversity mostly arose from richness/abundance differences suggests that, when compositional changes are neutrally driven, gains and losses in species and species abundances are the primary underlying mechanisms. Both habitat connectivity and quality were

**Fig. 5** Predicted partial effects of patch characteristics on local contributions to beta-diversity ( $LCBD_{Tot}$ ) and its species replacement ( $LCBD_{Repl}$ ) and richness/abundance difference components ( $LCBD_{AbDiff}$ ) for step time 600. The solid line corresponds to the average predictions across the 50 landscape replicates and the shaded areas to the mean  $\pm$  SD intervals



involved in this process. The most unique assemblages in terms of species abundances were actually those experiencing the highest mortalities and the greatest isolation, suggesting a critical role of ecological drift and immigration rates.

Species replacement never dominated in our landscapes, which is consistent with the usual assumption that this diversity component tends to assume great importance when environmental filtering is at play (Qian and Ricklefs 2007; Baselga 2010; Leprieur et al. 2011). Yet, its non-negligible contribution to total beta-diversity in our simulations (i.e., about 25%) also demonstrates that neutral processes can still cause spatial variation in species pools. Possible neutral drivers of species replacement include both dispersal and historical constraints (Morlon et al. 2008; Wang et al. 2020). While historical effects, resulting from the random settlement of species in the habitat

patches, likely played a role, the impact of dispersal constraints may have been minor. This is supported by the lack of significant effects of patch connectivity on  $LCBD_{Repl}$ . In the simulated landscapes, individuals were able to move between all pairs of patches. The level of dispersal constraint may thus not have been strong enough to provoke species replacement. The assemblages with the highest  $LCBD_{Repl}$  were those experiencing the lowest levels of mortality, and vice-versa. In other words, the sites with the most unique assemblages for harboring species absent in most other sites occurred preferentially under low mortality conditions. In the other sites, the mortality rates likely caused local extinctions and therefore resulted in loss of rare species from these sites if, as found by previous simulation studies (Fodelianakis et al. 2021), rare species were more prone to the effects of ecological drift. Considering that low-abundance species are

particularly vulnerable to the effects of drift (Gravel et al. 2011; Siqueira et al. 2020), it is highly probable that the vulnerability of rare species to drift was increased by their low abundances in our landscapes (mean  $\pm$  SD correlation between species abundance and occurrence for time-step 600 =  $0.81 \pm 0.05$ ).

#### The different aspects of community uniqueness and conservation implications

Community uniqueness in terms of species replacement and richness/abundance differences displayed opposite responses to both community and patch attributes, indicating that these two key community uniqueness aspects convey different information. These findings confirm that species replacement and richness/abundance call for different conservation strategies (Baselga 2010; Benito et al. 2020; Zhao et al. 2021) and stress the importance of partitioning LCBD into its components to propose adequate conservation strategies (Ruhí et al. 2017; Hill et al. 2021).

Community uniqueness in species replacement is of particular relevance for the preservation of regional biodiversity (Hill et al. 2021; Zhao et al. 2021) as it indicates assemblages with species that are absent in most other sites. Our study reinforces the significance of this parameter in conservation efforts. Assemblages with high community uniqueness in species replacement were characterized by high species richness and abundances of rare species, and were found in habitats less susceptible to ecological drift. In the last time steps, the most unique assemblages in species replacement were actually found in habitats that provided optimal conditions for population growth and low competition for space, as indicated by the detected effects of patch mortality and size. These findings confirm that large patches can include species that are absent in small patches (Diamond 1976; Fletcher and Fortin 2018), highlighting the importance of protecting and managing such habitats. However, this does not preclude the need for preserving small patches. Our study exclusively focused on single-patch contributions to beta-diversity and did not consider cumulative effects. Convincing evidence indicates that, contrary to what was initially postulated (Diamond 1975), several small patches usually hold more species than a few large patches of the same total area (Fahrig 2020; Riva and Fahrig 2022).

Future investigations are thus needed to evaluate whether small patches could also bring more community uniqueness than large patches totaling the same area. Besides, small patches may also play a critical role in overall ecosystem persistence (Tulloch et al. 2016).

In our simulated landscapes, assemblages with high uniqueness in terms of richness/abundance difference tended to identify assemblages in possible need of restoration. Unlike assemblages with high uniqueness in species replacement, assemblages with high uniqueness in richness/abundance differences had low richness and abundances of rare species. These assemblages tended to occur in isolated patches and in sites experiencing high mortalities, suggesting that they were under strong pressures from ecological drift. This finding stresses that high community uniqueness values do not necessarily identify sites of high conservation value for harboring unusual species combinations but can also refer to diversity-poor and possibly endangered communities (Legendre and De Cáceres 2013).

#### Effects of patch characteristics on community uniqueness

Understanding how patch characteristics influence community uniqueness is relevant to prioritize conservation in fragmented landscapes, but few empirical studies have addressed this issue (Dubois et al. 2020; Santos et al. 2021; Yan et al. 2021; Zhao et al. 2021; Lozada and Bertin 2022) and evidence to date has been inconsistent and contradictory. For instance, some studies found that small and poorly connected habitat patches were those presenting the most unique species assemblages (Zhao et al. 2021), while other studies did not (Yan et al. 2021; Lozada and Bertin 2022).

Here, we found a predominant impact of patch mortality on all community uniqueness aspects. While evidence has shown that habitat quality can alter communities by modifying species density (Wettstein and Schmid 1999) and richness (Delciellos et al. 2016), to our knowledge, no studies so far have explicitly investigated such impacts of habitat quality. Yet, community uniqueness has been associated with a wide range of environmental factors (Santos et al. 2021; Yao et al. 2021), associations that at least in some cases may have arisen from the influence of the

environmental factors on population and community growth. Our results demonstrate that habitat quality can affect both beta-diversity components, driving opposite responses. Good-quality environments (i.e., low mortality patches) promoted uniqueness in species replacement, likely due to the greater success of rare species in favorable environments. However, these same environments decreased the singularity of assemblages in terms of richness/abundance difference. More singular assemblages in richness/abundance difference thus occurred in poor quality environments, which was due to their atypically low levels of diversity. The fact that species mortality increased  $LCBD_{AbDiff}$  actually indicates that it altered species richness and abundance. As expected, high mortality was associated with the lowest levels of species richness and abundance (Appendix S1: Table S8). Taken together, these findings indicate that high-quality environments foster assemblages with high richness and abundances of rare species, underscoring the critical role of this habitat feature in promoting biodiversity and the importance of considering this aspect when developing conservation strategies.

In contrast to habitat quality, which impacted all aspects of beta-diversity, the effects of patch connectivity and size were more limited. Low patch connectivity was found to be associated with high  $LCBD_{AbDiff}$ , indicating that assemblages with low richness were more likely to occur in isolated sites where fewer species may have been able to arrive. On the other hand, large patch size was associated with a higher  $LCBD$  in species replacement in the last time steps, indicating that assemblages with higher richness and abundances of rare species were more likely to persist in large patches.

## Conclusions and future directions

In conclusion, our study sheds light on the mechanisms underlying beta-diversity and the spatial patterns of community uniqueness in fragmented landscapes. We found that  $LCBD_{Repl}$  and  $LCBD_{AbDiff}$  represent drastically different aspects of community uniqueness, often displaying opposite responses to ecological factors. Our results highlight the crucial role of habitat quality for beta-diversity, suggesting that it should be a primary focus of conservation efforts. However, the importance of patch attributes

may depend on a number of landscape characteristics. Since our simulated landscapes allowed for dispersal among all patches, they are well suited for investigating highly mobile organisms or analyzing small-scale systems. However, they may not adequately account for significant dispersal constraints in larger-scale studies or for organisms with limited dispersal abilities. Future research should thus evaluate the consistency of our results across a range of other spatial and ecological contexts, considering more restricted connectivity schemes and the effects of environmental filtering.

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**Data availability** All input files used to simulate communities and the output files produced for each landscape replicate and investigated time-step are already in the public domain. They are available at the figshare repository <https://doi.org/10.6084/m9.figshare.22581868> (Lozada et al. 2023). All the other data used in this study are provided in the Supporting Information file.

**Declarations**

**Competing interest** The authors declare they have no financial interests or conflict of interests.

**Ethical approval** Not applicable.

**Consent for publication** All authors contributed to this study and agree with the content of the manuscript for publication.

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